Exercises

Prof. Luca Cagliero, Dr. Giuseppe Gallipoli
Dipartimento di Automatica e Informatica
Politecnico di Torino

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Exercise 1

In the self-attention mechanism implemented in BERT

- *The attention score of a reference token is computed separately for each token in the sequence, including the reference token itself.*
- *The attention score of a reference token is computed separately for each token in the sequence, except for the reference token itself.*
- *The attention score of a reference token is computed by attending only part of the sequence according to the positional encoding.*
- *The attention score of special tokens (e.g.,* SEP*) is computed by attending only other special tokens (e.g.,* CLS*).*
- *None of the above.*

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Exercise 1 solution

In the self-attention mechanism implemented in BERT

- ***The attention score of a reference token is computed separately for each token in the sequence, including the reference token itself.***
- *The attention score of a reference token is computed separately for each token in the sequence, except for the reference token itself.*
- *The attention score of a reference token is computed by attending only part of the sequence according to the positional encoding.*
- *The attention score of special tokens (e.g.,* SEP*) is computed by attending only other special tokens (e.g.,* CLS*).*
- *None of the above.*

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Exercise 2 solution

Rule-based Named Entity Recognition is not enhanced by

- *Syntactic Parsing.*
- *POS tagging.*
- *Lexical databases.*
- *Domain-specific ontological models.*
- *Stemming.*

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Exercise 2 solution

Rule-based Named Entity Recognition is not enhanced by

- *Syntactic Parsing.*
- *POS tagging.*
- *Lexical databases.*
- *Domain-specific ontological models.*
- ***Stemming.***

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Exercise 3

Which of the following is a valid definition of n-gram?

- *A set of n sentences that are part of a document.*
- *A bag of words describing a document topic.*
- *A contiguous sequence of n characters that are part of a word.*
- *A set of n phonemes that are part of a phrase.*
- *None of the above.*

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Exercise 3 solution

Which of the following is a valid definition of n-gram?

- *A set of n sentences that are part of a document.*
- *A bag of words describing a document topic.*
- ***A contiguous sequence of n characters that are part of a word.***
- *A set of n phonemes that are part of a phrase.*
- *None of the above.*

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Exercise 3

Given the following SVD decomposition of a word-document matrix **A** with documents d_1, d_2, d_3 and words w_1, ..., w_{11}:

$$\mathbf{A} = \mathbf{U}\mathbf{S}\mathbf{V}^T$$

$$\mathbf{U} = \begin{bmatrix} -0.4201 & 0.0748 & -0.0460 \\ -0.2995 & -0.2001 & 0.4078 \\ -0.1206 & 0.2749 & -0.4538 \\ -0.1576 & -0.3046 & -0.2006 \\ -0.1206 & 0.2749 & -0.4538 \\ -0.2626 & 0.3749 & 0.1547 \\ -0.4201 & 0.0748 & -0.0460 \\ -0.4201 & 0.0748 & -0.0460 \\ -0.2626 & 0.3794 & 0.1547 \\ -0.3151 & -0.6093 & -0.4013 \\ -0.2995 & -0.2001 & 0.4078 \end{bmatrix}$$

$$\mathbf{S} = \begin{bmatrix} 4.0989 & 0.0000 & 0.0000 \\ 0.0000 & 2.3616 & 0.0000 \\ 0.0000 & 0.0000 & 1.2737 \end{bmatrix}$$

$$\mathbf{V} = \begin{bmatrix} -0.4945 & 0.6492 & -0.5780 \\ -0.6458 & -0.7194 & -0.2556 \\ -0.5817 & 0.2469 & 0.7750 \end{bmatrix}$$

$$\mathbf{V}^T = \begin{bmatrix} -0.4945 & -0.6458 & -0.5817 \\ 0.6492 & -0.7194 & 0.2469 \\ -0.5780 & -0.2556 & 0.7750 \end{bmatrix}$$

Find the word-level description of each topic.

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Exercise 3 – Solution

Before computing the word-level description of each topic, we check if we can prune the set of topics.

$$\mathbf{S} = \begin{bmatrix} 4.0989 & 0.0000 & 0.0000 \\ 0.0000 & 2.3616 & 0.0000 \\ 0.0000 & 0.0000 & 1.2737 \end{bmatrix}$$

Highest eigenvalue $\sigma_1 = 4.0989$

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Exercise 3 – Solution

Before computing the word-level description of each topic, we check if we can prune the set of topics.

$$\mathbf{S} = \begin{bmatrix} 4.0989 & 0.0000 & 0.0000 \\ 0.0000 & 2.3616 & 0.0000 \\ 0.0000 & 0.0000 & 1.2737 \end{bmatrix}$$

Highest eigenvalue $\sigma_1 = 4.0989$

$$\sigma_2 > \frac{\sigma_1}{2} \rightarrow 2.3616 > \frac{4.0989}{2} \ ?$$

Yes, we keep it.

$$\sigma_3 > \frac{\sigma_1}{2} \rightarrow 1.2737 > \frac{4.0989}{2} \ ?$$

No, we prune it.

Resulting $\mathbf{S} = \begin{bmatrix} 4.0989 & 0.0000 & 0.0000 \\ 0.0000 & 2.3616 & 0.0000 \\ 0.0000 & 0.0000 & \boxed{0.0000} \end{bmatrix}$

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Exercise 3 – Solution

$$\mathbf{U} = \begin{bmatrix} -0.4201 & 0.0748 & -0.0460 \\ -0.2995 & -0.2001 & 0.4078 \\ -0.1206 & 0.2749 & -0.4538 \\ -0.1576 & -0.3046 & -0.2006 \\ -0.1206 & 0.2749 & -0.4538 \\ -0.2626 & 0.3749 & 0.1547 \\ -0.4201 & 0.0748 & -0.0460 \\ -0.4201 & 0.0748 & -0.0460 \\ -0.2626 & 0.3794 & 0.1547 \\ -0.3151 & -0.6093 & -0.4013 \\ -0.2995 & -0.2001 & 0.4078 \end{bmatrix}$$

$$\mathbf{S} = \begin{bmatrix} 4.0989 & 0.0000 & 0.0000 \\ 0.0000 & 2.3616 & 0.0000 \\ 0.0000 & 0.0000 & 0.0000 \end{bmatrix}$$

Word relevance scores of Topic #1:

- $s_{1,1}$ =

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Exercise 3 – Solution

$$\mathbf{U} = \begin{bmatrix} -0.4201 & 0.0748 & -0.0460 \\ -0.2995 & -0.2001 & 0.4078 \\ -0.1206 & 0.2749 & -0.4538 \\ -0.1576 & -0.3046 & -0.2006 \\ -0.1206 & 0.2749 & -0.4538 \\ -0.2626 & 0.3749 & 0.1547 \\ -0.4201 & 0.0748 & -0.0460 \\ -0.4201 & 0.0748 & -0.0460 \\ -0.2626 & 0.3794 & 0.1547 \\ -0.3151 & -0.6093 & -0.4013 \\ -0.2995 & -0.2001 & 0.4078 \end{bmatrix} \begin{array}{l} w_1 \\ \\ \\ \\ \\ \\ \\ \\ \\ \\ \\ \end{array}$$

$$\mathbf{S} = \begin{array}{c} \begin{array}{ccc} t_1 & & \end{array} \\ \begin{bmatrix} 4.0989 & 0.0000 & 0.0000 \\ 0.0000 & 2.3616 & 0.0000 \\ 0.0000 & 0.0000 & 0.0000 \end{bmatrix} \end{array}$$

Word relevance scores of Topic #1:

- $s_{1,1} = w_1 \times t_1 =$ -0.4201 x 4.0989 = -1.722

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Exercise 3 – Solution

$$\mathbf{U} = \left[\begin{array}{rrr} -0.4201 & 0.0748 & -0.0460 \\ -0.2995 & -0.2001 & 0.4078 \\ -0.1206 & 0.2749 & -0.4538 \\ -0.1576 & -0.3046 & -0.2006 \\ -0.1206 & 0.2749 & -0.4538 \\ -0.2626 & 0.3749 & 0.1547 \\ -0.4201 & 0.0748 & -0.0460 \\ -0.4201 & 0.0748 & -0.0460 \\ -0.2626 & 0.3794 & 0.1547 \\ -0.3151 & -0.6093 & -0.4013 \\ -0.2995 & -0.2001 & 0.4078 \end{array}\right] \begin{array}{l} \\ w_2 \\ \\ \\ \\ \\ \\ \\ \\ \\ \\ \end{array}$$

$$\mathbf{S} = \begin{array}{c} \begin{array}{ccc} t_1 & & \end{array} \\ \begin{bmatrix} 4.0989 & 0.0000 & 0.0000 \\ 0.0000 & 2.3616 & 0.0000 \\ 0.0000 & 0.0000 & 0.0000 \end{bmatrix} \end{array}$$

Word relevance scores of Topic #1:

- $s_{1,1}$ = w_1 x t_1 = -0.4201 x 4.0989 = -1.722
- $s_{2,1}$ = w_2 x t_1 = -0.2995 x 4.0989 = -1.228

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Exercise 3 – Solution

$$\mathbf{U} = \begin{array}{cc} \begin{bmatrix} -0.4201 & 0.0748 & -0.0460 \\ -0.2995 & -0.2001 & 0.4078 \\ -0.1206 & 0.2749 & -0.4538 \\ -0.1576 & -0.3046 & -0.2006 \\ -0.1206 & 0.2749 & -0.4538 \\ -0.2626 & 0.3749 & 0.1547 \\ -0.4201 & 0.0748 & -0.0460 \\ -0.4201 & 0.0748 & -0.0460 \\ -0.2626 & 0.3794 & 0.1547 \\ -0.3151 & -0.6093 & -0.4013 \\ -0.2995 & -0.2001 & 0.4078 \end{bmatrix} & \begin{array}{c} \\ \\ w_3 \\ \\ \\ \\ \\ \\ \\ \\ \\ \end{array} \end{array}$$

$$\mathbf{S} = \begin{array}{c} \begin{array}{ccc} t_1 & & \end{array} \\ \begin{bmatrix} 4.0989 & 0.0000 & 0.0000 \\ 0.0000 & 2.3616 & 0.0000 \\ 0.0000 & 0.0000 & 0.0000 \end{bmatrix} \end{array}$$

Word relevance scores of Topic #1:

- $s_{1,1} = w_1 \times t_1 =$ -0.4201 x 4.0989 = -1.722
- $s_{2,1} = w_2 \times t_1 =$ -0.2995 x 4.0989 = -1.228
- $s_{3,1} = w_3 \times t_1 =$ -0.1206 x 4.0989 = -0.494

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Exercise 3 – Solution

$$\mathbf{U} = \begin{bmatrix} -0.4201 & 0.0748 & -0.0460 \\ -0.2995 & -0.2001 & 0.4078 \\ -0.1206 & 0.2749 & -0.4538 \\ -0.1576 & -0.3046 & -0.2006 \\ -0.1206 & 0.2749 & -0.4538 \\ -0.2626 & 0.3749 & 0.1547 \\ -0.4201 & 0.0748 & -0.0460 \\ -0.4201 & 0.0748 & -0.0460 \\ -0.2626 & 0.3794 & 0.1547 \\ -0.3151 & -0.6093 & -0.4013 \\ -0.2995 & -0.2001 & 0.4078 \end{bmatrix} \qquad \mathbf{S} = \begin{bmatrix} 4.0989 & 0.0000 & 0.0000 \\ 0.0000 & 2.3616 & 0.0000 \\ 0.0000 & 0.0000 & 0.0000 \end{bmatrix}$$

Word relevance scores of Topic #1:

- $s_{1,1}$ = w_1 x t_1 = -0.4201 x 4.0989 = -1.722
- $s_{2,1}$ = w_2 x t_1 = -0.2995 x 4.0989 = -1.228
- $s_{3,1}$ = w_3 x t_1 = -0.1206 x 4.0989 = **-0.494**
- $s_{4,1}$ = w_4 x t_1 = -0.1576 x 4.0989 = -0.646
- $s_{5,1}$ = w_5 x t_1 = -0.1206 x 4.0989 = **-0.494**
- $s_{6,1}$ = w_6 x t_1 = -0.2626 x 4.0989 = -1.076
- $s_{7,1}$ = w_7 x t_1 = -0.4201 x 4.0989 = -1.722
- $s_{8,1}$ = w_8 x t_1 = -0.4201 x 4.0989 = -1.722
- $s_{9,1}$ = w_9 x t_1 = -0.2626 x 4.0989 = -1.076
- $s_{10,1}$ = w_{10} x t_1 = -0.3151 x 4.0989 = -1.291
- $s_{11,1}$ = w_{11} x t_1 = -0.2995 x 4.0989 = -1.228

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Exercise 3 – Solution

$$\mathbf{U} = \begin{bmatrix} -0.4201 & 0.0748 & -0.0460 \\ -0.2995 & -0.2001 & 0.4078 \\ -0.1206 & 0.2749 & -0.4538 \\ -0.1576 & -0.3046 & -0.2006 \\ -0.1206 & 0.2749 & -0.4538 \\ -0.2626 & 0.3749 & 0.1547 \\ -0.4201 & 0.0748 & -0.0460 \\ -0.4201 & 0.0748 & -0.0460 \\ -0.2626 & 0.3794 & 0.1547 \\ -0.3151 & -0.6093 & -0.4013 \\ -0.2995 & -0.2001 & 0.4078 \end{bmatrix}$$

$$\mathbf{S} = \begin{bmatrix} 4.0989 & 0.0000 & 0.0000 \\ 0.0000 & 2.3616 & 0.0000 \\ 0.0000 & 0.0000 & 0.0000 \end{bmatrix}$$

Word relevance scores of Topic #2:

- $s_{1,2}$ =

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Exercise 3 – Solution

$$\mathbf{U} = \begin{array}{c} \begin{bmatrix} -0.4201 & 0.0748 & -0.0460 \\ -0.2995 & -0.2001 & 0.4078 \\ -0.1206 & 0.2749 & -0.4538 \\ -0.1576 & -0.3046 & -0.2006 \\ -0.1206 & 0.2749 & -0.4538 \\ -0.2626 & 0.3749 & 0.1547 \\ -0.4201 & 0.0748 & -0.0460 \\ -0.4201 & 0.0748 & -0.0460 \\ -0.2626 & 0.3794 & 0.1547 \\ -0.3151 & -0.6093 & -0.4013 \\ -0.2995 & -0.2001 & 0.4078 \end{bmatrix} \end{array} \begin{array}{l} w_1 \\ \\ \\ \\ \\ \\ \\ \\ \\ \\ \\ \end{array}$$

$$\mathbf{S} = \begin{array}{c} \begin{array}{ccc} & t_2 & \end{array} \\ \begin{bmatrix} 4.0989 & \boxed{0.0000} & 0.0000 \\ 0.0000 & \boxed{2.3616} & 0.0000 \\ 0.0000 & \boxed{0.0000} & 0.0000 \end{bmatrix} \end{array}$$

Word relevance scores of Topic #2:

- $s_{1,2} = w_1 \times t_2 = 0.0748 \times 2.3616 = 0.177$

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Exercise 3 – Solution

$$\mathbf{U} = \begin{bmatrix} -0.4201 & 0.0748 & -0.0460 \\ \boxed{-0.2995 \quad -0.2001 \quad 0.4078} \\ -0.1206 & 0.2749 & -0.4538 \\ -0.1576 & -0.3046 & -0.2006 \\ -0.1206 & 0.2749 & -0.4538 \\ -0.2626 & 0.3749 & 0.1547 \\ -0.4201 & 0.0748 & -0.0460 \\ -0.4201 & 0.0748 & -0.0460 \\ -0.2626 & 0.3794 & 0.1547 \\ -0.3151 & -0.6093 & -0.4013 \\ -0.2995 & -0.2001 & 0.4078 \end{bmatrix} \begin{array}{l} \\ w_2 \\ \\ \\ \\ \\ \\ \\ \\ \\ \\ \end{array}$$

$$\mathbf{S} = \begin{array}{c} \begin{array}{ccc} & t_2 & \end{array} \\ \begin{bmatrix} 4.0989 & \boxed{0.0000} & 0.0000 \\ 0.0000 & \boxed{2.3616} & 0.0000 \\ 0.0000 & \boxed{0.0000} & 0.0000 \end{bmatrix} \end{array}$$

Word relevance scores of Topic #2:

- $s_{1,2} = w_1 \times t_2 = 0.0748 \times 2.3616 = 0.177$
- $s_{2,2} = w_2 \times t_2 = -0.2001 \times 2.3616 = -0.472$

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Exercise 2 – Solution

$$\mathbf{U} = \begin{array}{cc} \begin{bmatrix} -0.4201 & 0.0748 & -0.0460 \\ -0.2995 & -0.2001 & 0.4078 \\ -0.1206 & 0.2749 & -0.4538 \\ -0.1576 & -0.3046 & -0.2006 \\ -0.1206 & 0.2749 & -0.4538 \\ -0.2626 & 0.3749 & 0.1547 \\ -0.4201 & 0.0748 & -0.0460 \\ -0.4201 & 0.0748 & -0.0460 \\ -0.2626 & 0.3794 & 0.1547 \\ -0.3151 & -0.6093 & -0.4013 \\ -0.2995 & -0.2001 & 0.4078 \end{bmatrix} & \begin{array}{c} \\ \\ w_3 \\ \\ \\ \\ \\ \\ \\ \\ \\ \end{array} \end{array}$$

$$\mathbf{S} = \begin{array}{c} \begin{array}{ccc} & t_2 & \end{array} \\ \begin{bmatrix} 4.0989 & 0.0000 & 0.0000 \\ 0.0000 & 2.3616 & 0.0000 \\ 0.0000 & 0.0000 & 0.0000 \end{bmatrix} \end{array}$$

Word relevance scores of Topic #2:

- $s_{1,2} = w_1 \times t_2 = 0.0748 \times 2.3616 = 0.177$
- $s_{2,2} = w_2 \times t_2 = -0.2001 \times 2.3616 = -0.472$
- $s_{3,2} = w_3 \times t_2 = 0.2749 \times 2.3616 = 0.649$

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Exercise 3 – Solution

$$\mathbf{U} = \begin{bmatrix} -0.4201 & 0.0748 & -0.0460 \\ -0.2995 & -0.2001 & 0.4078 \\ -0.1206 & 0.2749 & -0.4538 \\ -0.1576 & -0.3046 & -0.2006 \\ -0.1206 & 0.2749 & -0.4538 \\ -0.2626 & 0.3749 & 0.1547 \\ -0.4201 & 0.0748 & -0.0460 \\ -0.4201 & 0.0748 & -0.0460 \\ -0.2626 & 0.3794 & 0.1547 \\ -0.3151 & -0.6093 & -0.4013 \\ -0.2995 & -0.2001 & 0.4078 \end{bmatrix}$$

$$\mathbf{S} = \begin{bmatrix} 4.0989 & 0.0000 & 0.0000 \\ 0.0000 & 2.3616 & 0.0000 \\ 0.0000 & 0.0000 & 0.0000 \end{bmatrix}$$

Word relevance scores of Topic #2:

- $s_{1,2} = w_1 \times t_2 = 0.0748 \times 2.3616 = 0.177$
- $s_{2,2} = w_2 \times t_2 = -0.2001 \times 2.3616 = -0.472$
- $s_{3,2} = w_3 \times t_2 = 0.2749 \times 2.3616 = 0.649$
- $s_{4,2} = w_4 \times t_2 = -0.3046 \times 2.3616 = -0.719$
- $s_{5,2} = w_5 \times t_2 = 0.2749 \times 2.3616 = 0.649$
- $s_{6,2} = w_6 \times t_2 = 0.3749 \times 2.3616 =$ **0.885**
- $s_{7,2} = w_7 \times t_2 = 0.0748 \times 2.3616 = 0.177$
- $s_{8,2} = w_8 \times t_2 = 0.0748 \times 2.3616 = 0.177$
- $s_{9,2} = w_9 \times t_2 = 0.3794 \times 2.3616 =$ **0.896**
- $s_{10,2} = w_{10} \times t_2 = -0.6093 \times 2.3616 = -1.439$
- $s_{11,2} = w_{11} \times t_2 = -0.2001 \times 2.3616 = -0.472$

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Exercise 4

The PoliTo company need to identify critical reviews crawled from the web. Reviews are not necessarily written in English. Let us consider a list of text snippets containing the reviews as input. Address the following points:

1. Define an NLP pipeline leveraging pre-trained models. Let us suppose that no annotations are provided at this stage.
2. In view of the positive feedback received from your company supervisor, you are now allowed to ask for 5k annotations. Describe the annotations you are willing to collect and how they can be used to improve the performance of your model.

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Exercise 4 – Solution (part 1)

Main pipeline steps (example)

1. Language identification.
2. Pre-trained machine translation model applied to non-target language reviews.
3. Train sentence embedding model on different sentiment-aware datasets or use a pre-trained model to map reviews into vector embeddings.
4. Apply clustering to distinguish the 2 clusters.
5. Manually inspect the clusters to decide which one contains the critical reviews.

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Exercise 4 – Solution (part 2)

Main pipeline steps (example)

1. Sampling from different clusters to choose 5k data sentences.
2. Let human annotator label the reviews in the 5k sampled data as critical or non-critical.
3. Train a supervised classification model to perform binary classification.
4. Use train/validation/test (80%/10%/10%) split. Validation set is used to stop model training once overfitting is detected.
5. Assess the model performance.

Optional (prior to step 2): Get a higher amount of samples from near cluster borders.

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Exercise 5

Given a financial news you want to detect the relations between different text spans. Specifically, the goal is to "identify and extract", within a sentence or a longer text block contained in the news content, the causal elements and the consequential ones. Assume that they consist of distinct cause-effect pairs.

Example:

Original news content: *Zhao found himself 60 million yuan indebted after losing 9,000 BTC in a single day (February 10, 2014)*

Extracted cause: *losing 9,000 BTC in a single day (February 10, 2014)*

Extracted effect: *Zhao found himself 60 million yuan indebted*

Assume that a ground-truth consisting of 1000 <*content, cause, effect*> triples are given.

1. Design a complete NLP pipeline for accomplishing the task. For each pipeline step clarify the goal, the algorithms used, and the main settings.

2. Describe the metrics used to evaluate the system performance and explain their meaning.

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Exercise 5.1 solution

- Language identification and machine translation (if need be)
- Selection of a machine learning model able to classify single tokens. In this case, it could be useful to select a pre-trained encoder model (e.g., BERT)
- Text tokenization using a tokenizer selected according to the previous step.
- Training/Fine-tuning a token classification model (similar to NER).
- Examples of annotations used in the training phase:
  - B-CAUSE: begin token for the clause
  - I-CAUSE: token inside a clause (including final token)
  - B-EFFECT: begin token for the effect
  - I-EFFECT: token inside a effect (including final token)
  - OTHER: token outside clause and effect
- Running the fine-tuned model on inference model (e.g., BERT) to detect cause/effect snippets.

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Exercise 5.2 solution

The following two metrics can be used to evaluate the proposed system

- ROUGE-based scores can be used to identify the overlap between the system and reference text snippets.
- EXACT/ APPROXIMATE MATCH can be used to compute the fraction of cause/effect that are correctly identified.

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- Copyright licence
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- Acknowledgements
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- Affiliation
  - The author and his staff are currently members of the Database and Data Mining Group at Dipartimento di Automatica e Informatica (Politecnico di Torino) and of the SmartData interdepartmental centre
    - https://dbdmg.polito.it
    - https://smartdata.polito.it

Thank you!

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